The Islamia University of Bahawalpur

Bahawalpur Baghdad Campus

Final Year Project Proposal

Program: ☐ BSSE ☑ BSIT ☐ BSAI

Project Title: Aegis-Proxy: Adversarial Security Gateway for Enterprise AI Systems

Nature of Project: ☑ Development ☐ Research

Area of Specialization:

☑ Artificial Intelligence (AI)	☐ Game Development	☐ Internet of Things (IoT)
☑ Machine Learning	☐ Mobile App Development	☐ Blockchain
☐ Data Science and Analytics	☑ Web Development	☑ Cybersecurity
☐ Image Processing	☐ Desktop Application	
☑ Natural Language Processing (NLP)	☐ Other (Specify)_____	

Introduction:

AI-powered applications are increasingly used for information retrieval, content generation, and business tasks. However, these systems can be targeted through prompt injection and jailbreak attempts, where crafted inputs try to override intended instructions or make a model disclose information it should not reveal. Aegis-Proxy is proposed as a security gateway between an application and its AI model. It will inspect incoming prompts using rule-based checks and machine-learning techniques, calculate a risk score, and apply configurable policies to allow, flag, or block suspicious requests. The project is relevant because it adds a practical security layer to AI integrations and provides visibility into detected threats through audit logs and an administrative dashboard.

Problem Statement:

AI models may respond to maliciously crafted prompts that attempt to bypass safety instructions, reveal sensitive information, or misuse an application's intended behavior. Many application integrations rely mainly on the model's own safeguards and do not provide a separate, configurable gateway for inspecting requests before they reach the model. This creates several challenges: (1) prompt injection and jailbreak attempts may be difficult to identify consistently; (2) simple keyword rules can miss varied wording, while overly strict rules may flag harmless prompts; (3) developers may lack risk-based allow, flag, and block controls; and (4) security events may not be centrally logged for review and evaluation. Aegis-Proxy will address these gaps with combined rule-based and ML detection, risk scoring, policy-based decisions, and audit logging. The prototype will be evaluated on a prepared dataset; it will not claim to detect every possible attack.

Project Goal / Objectives:

(List the main goals and measurable objectives.)

1. Design and implement a FastAPI gateway that receives prompts and securely forwards permitted requests to an AI model API.
2. Combine configurable rule-based checks with a machine-learning classifier to identify potential prompt injection and jailbreak attempts.
3. Assign risk scores and apply configurable allow, flag, or block policies before forwarding requests.
4. Detect and redact selected sensitive information where configured, and record security events in PostgreSQL for audit and review.
5. Build an administrative dashboard for reviewing detections, risk scores, policy decisions, and security logs, with authentication and role-based access control.
6. Evaluate the prototype using precision, recall, F1-score, false-positive rate, and response latency.

Application Architecture / Technology Stack / Tools / Hardware:

(Mention the system architecture, platforms, frameworks, and tools to be used.)
Example

Architecture: Modular web application with a FastAPI security gateway between the client application and an external AI model API; Next.js dashboard communicates with the backend through REST APIs.

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS.

Frameworks / ML: FastAPI, Next.js, React.js, Tailwind CSS, PyTorch, Hugging Face Transformers.

Database / tools: PostgreSQL, Redis (if required for caching or rate limits), Pytest, Docker, Git, OpenAPI/Swagger.

Hardware: Standard development laptop/PC with internet access; no dedicated sensor hardware is required.

Project Group Members:

Sr.#	Reg. #	Student Name	Phone #	Signature
(1)	F23BINFT1M01002	Zainab Fatima	03338594955	

References (if any):

1. OWASP Foundation. OWASP Top 10 for Large Language Model Applications. https://owasp.org/www-project-top-10-for-large-language-model-applications/

2. Greshake, K. et al. (2023). “Not what you’ve signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection.” arXiv:2302.12173. https://arxiv.org/abs/2302.12173

3. Project implementation and evaluation will also use relevant documentation for FastAPI, Hugging Face Transformers, and the selected AI model API.

Project Supervisor:

Name: Dr. Rana Nazir Ahmed

Supervisor Acceptance Signature: _____

Head of Department (HoD): _____